

SURIN, Thailand

WMO: 484320

Lat: 14.874N

Lon: 103.499E

Elev: 147

StdP: 99.57

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 15.3	(c) 16.8	(d) 9.9	(e) 7.7	(f) 19.4	(g) 11.3	(h) 8.5	(i) 20.4	(j) 7.0	(k) 24.6	(l) 6.1	(m) 24.9	(n) 1.2	(o) 30	(p) 0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 9.8	(c) 37.5	(d) 25.2	(e) 36.5	(f) 25.2	(g) 35.5	(h) 25.2	(i) 27.1	(j) 32.8	(k) 26.8	(l) 32.3	(m) 26.6	(n) 31.9	(o) 1.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.5	28.8	25.6	21.2	28.7	25.3	20.9	28.5	86.5	32.9	85.3	32.5	84.3	32.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.3	(b) 4.5	(c) 3.9	(e) 12.8	(f) 38.8	(g) 1.4	(h) 0.8	(i) 11.7	(j) 39.4	(k) 10.9	(l) 39.9	(m) 10.1	(n) 40.4	(o) 9.1	(p) 41.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			10.8	28.0	1.6	0.6	9.7	28.4	8.8	28.8	7.9	29.1	6.7	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	24.8	26.9	29.2	30.4	29.8	29.3	28.6	28.3	27.9	27.4	26.2	24.5
	DBStd	2.66	2.56	2.85	2.44	1.93	1.57	1.25	1.22	1.11	1.17	1.29	2.11	2.65
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6482	459	473	595	611	613	578	575	568	536	538	487	449
	CDD18.3	3441	201	239	337	361	354	328	317	310	286	280	237	191
	CDH23.3	37992	1904	2692	4254	4677	4294	3821	3454	3303	2893	2767	2219	1715
	CDH26.7	16578	778	1308	2280	2532	2049	1658	1334	1223	977	956	847	636
(14) Wind	WSAvg	1.2	1.1	1.1	1.1	1.1	1.1	1.2	1.3	1.2	0.9	1.1	1.3	1.5
(15) Precipitation	PrecAvg	1289	2	10	23	83	173	166	185	209	284	127	25	3
	PrecMax	1626	16	63	52	226	451	295	343	369	519	319	81	27
	PrecMin	985	0	0	0	6	45	92	84	66	133	36	0	0
	PrecStd	180	4	16	16	61	106	62	68	85	119	79	26	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	36.7	38.3	39.2	37.9	36.0	35.1	34.2	33.7	33.4	34.2	34.0
		MCWB	23.1	23.9	24.4	25.4	26.4	26.1	26.1	26.3	26.4	25.7	25.1	23.7
	2%	DB	33.3	35.5	37.1	37.8	36.3	34.8	33.7	33.1	32.7	32.4	33.1	32.7
		MCWB	22.9	23.8	24.4	25.4	26.2	26.1	26.0	26.2	26.3	25.4	24.6	23.4
	5%	DB	32.2	34.5	36.1	36.7	35.1	33.8	32.8	32.4	31.8	31.6	32.1	31.6
		MCWB	22.6	23.7	24.5	25.5	26.2	26.2	25.9	26.1	26.1	25.2	24.3	22.9
	10%	DB	30.9	33.2	35.0	35.4	33.8	32.8	31.9	31.6	31.0	30.9	31.0	30.2
		MCWB	22.2	23.6	24.6	25.5	26.1	26.0	25.8	25.9	25.9	25.0	23.7	22.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.9	26.5	27.2	27.5	27.3	27.1	27.2	27.2	26.8	26.2	25.3
		MCDB	31.5	33.4	34.7	35.2	34.0	32.9	32.2	31.8	31.5	30.9	31.3	31.1
	2%	WB	24.4	25.3	25.9	26.7	27.1	26.9	26.6	26.7	26.7	26.3	25.7	24.5
		MCDB	30.4	32.5	34.0	34.4	33.3	32.4	31.5	31.2	30.8	30.2	30.6	30.3
	5%	WB	23.7	24.8	25.5	26.2	26.7	26.6	26.3	26.5	26.5	26.0	25.2	23.8
		MCDB	29.7	31.8	33.3	33.7	32.6	31.9	31.0	30.8	30.4	29.8	29.7	29.5
	10%	WB	22.9	24.3	25.1	26.0	26.4	26.3	26.1	26.2	26.2	25.7	24.8	23.1
		MCDB	29.2	30.8	32.7	33.2	32.1	31.2	30.7	30.3	29.9	29.2	29.0	28.9
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	12.0	11.1	9.8	8.2	7.4	6.9	6.7	6.3	7.0	9.0	10.5
		MCDBR	12.3	12.4	12.0	11.0	9.7	8.5	8.0	7.6	7.4	7.8	9.7	11.0
		MCWBR	3.8	3.3	2.9	2.4	2.3	2.1	2.2	2.2	2.4	2.4	2.8	3.4
	5% WB	MCDBR	10.8	10.7	10.7	9.9	8.6	7.8	7.3	7.1	6.9	6.9	8.6	10.0
		MCWBR	3.6	3.1	3.0	2.7	2.6	2.4	2.4	2.4	2.5	2.4	2.6	3.3
(40) Clear-Sky Solar Irradiance	taub		0.498	0.612	0.663	0.600	0.500	0.467	0.460	0.463	0.496	0.523	0.475	0.465
	taud		2.033	1.758	1.688	1.872	2.191	2.292	2.309	2.292	2.186	2.084	2.198	2.166
	Ebn at Noon		793	719	697	743	808	827	834	841	816	784	811	812
	Edh at Noon		166	228	250	208	149	133	131	135	150	162	140	142
(44) All-Sky Solar Radiation	RadAvg		5.01	5.25	5.38	5.75	5.68	5.36	5.08	4.90	4.68	4.94	4.95	4.84
	RadStd		0.27	0.22	0.20	0.23	0.30	0.20	0.32	0.30	0.31	0.33	0.23	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.31	N/A	N/A	N/A	+0.22	+0.12	N/A	N/A	+129
(47) Regional (18 neighbors)		+0.26	N/A	N/A	N/A	+0.25	+0.22	N/A	N/A	+103

Nomenclature: See separate page